

STAT3006 Worksheet 03

1. Bayesian Linear Regression with Unknown Variance

Consider the linear regression model

$$y_i = \mathbf{x}_i^T \boldsymbol{\theta} + \epsilon_i, \quad \epsilon_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$$

for observations $i = 1, \dots, n$ where $\mathbf{x}_i \in \mathbb{R}^p$ are the predictors and $\boldsymbol{\theta} \in \mathbb{R}^p$ are regression coefficients. In matrix form,

$$\mathbf{y} = \mathbf{X}\boldsymbol{\theta} + \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_n)$$

We assume both $\boldsymbol{\theta}$ and σ^2 are unknown and want to compute the posterior

$$\mathbb{P}(\boldsymbol{\theta}, \sigma^2 | \mathbf{X}, \mathbf{y})$$

(a) **Likelihood.** Show that

$$\mathbf{y} | \mathbf{X}, \boldsymbol{\theta}, \sigma^2 \sim \mathcal{N}(\mathbf{X}\boldsymbol{\theta}, \sigma^2 \mathbf{I}_n)$$

Write the likelihood $\mathbb{P}(\mathbf{y} | \mathbf{X}, \boldsymbol{\theta}, \sigma^2)$.

Answer: Since $\mathbf{y} = \mathbf{X}\boldsymbol{\theta} + \boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_n)$ is just a linear shift of $\boldsymbol{\epsilon}$ by the constant $\mathbf{X}\boldsymbol{\theta}$, we have

$$\mathbf{y} | \mathbf{X}, \boldsymbol{\theta}, \sigma^2 \sim \mathcal{N}(\mathbf{X}\boldsymbol{\theta}, \sigma^2 \mathbf{I}_n)$$

The likelihood is then given by

$$\mathbb{P}(\mathbf{y} | \mathbf{X}\boldsymbol{\theta}, \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp \left\{ \frac{-1}{2\sigma^2} (\mathbf{y} - \mathbf{X}\boldsymbol{\theta})^T (\mathbf{y} - \mathbf{X}\boldsymbol{\theta}) \right\}$$

(b) **Prior.** Assume the conjugate prior

$$\boldsymbol{\theta} | \sigma^2 \sim \mathcal{N}(\boldsymbol{\mu}_0, \sigma^2 \Sigma_0), \quad \sigma^2 \sim \text{Inv-Gamma}(\alpha_0, \beta_0)$$

Write the joint prior.

Answer: By Bayes Rule,

$$\begin{aligned}
\mathbb{P}(\boldsymbol{\theta}|\sigma^2) &= \frac{\mathbb{P}(\boldsymbol{\theta}, \sigma^2)}{\mathbb{P}(\sigma^2)} \\
\mathbb{P}(\boldsymbol{\theta}, \sigma^2) &= \mathbb{P}(\boldsymbol{\theta}|\sigma^2)\mathbb{P}(\sigma^2) \\
&= (2\pi\sigma^2)^{-p/2} |\Sigma_0|^{-1/2} \exp \left\{ \frac{-1}{2\sigma^2} (\boldsymbol{\theta} - \boldsymbol{\mu}_0)^T \Sigma_0^{-1} (\boldsymbol{\theta} - \boldsymbol{\mu}_0) \right\} \\
&\quad \cdot \frac{\beta_0^{\alpha_0}}{\Gamma(\alpha_0)} \sigma^{-2(\alpha_0+1)} \exp \left\{ \frac{-\beta_0}{\sigma^2} \right\} \\
&= \beta_0^{\alpha_0} (2\pi)^{-p/2} (\sigma^2)^{-p/2-\alpha_0-1} |\Sigma_0|^{-1/2} (\Gamma(\alpha_0))^{-1} \\
&\quad \cdot \exp \left\{ \frac{-1}{2\sigma^2} \left((\boldsymbol{\theta} - \boldsymbol{\mu}_0)^T \Sigma_0^{-1} (\boldsymbol{\theta} - \boldsymbol{\mu}_0) + 2\beta_0 \right) \right\}
\end{aligned}$$

(c) **Posterior (up to proportionality).** Using Bayes' rule, write the posterior distribution up to proportionality

Answer: As seen in lectures, the posterior is proportional to the likelihood multiplied by the prior. That is,

$$\mathbb{P}(\boldsymbol{\theta}, \sigma^2 | \mathbf{X}, \mathbf{y}) \propto \mathbb{P}(\mathbf{y} | \mathbf{X}, \boldsymbol{\theta}, \sigma^2) \cdot \mathbb{P}(\boldsymbol{\theta} | \sigma^2) \cdot \mathbb{P}(\sigma^2)$$

We can adopt the answer from part (c) and drop the constant terms that don't involve $\boldsymbol{\theta}$ or σ^2 as we are concerned with proportionality. This gives us

$$\begin{aligned}
\mathbb{P}(\boldsymbol{\theta}, \sigma^2 | \mathbf{X}, \mathbf{y}) &\propto (\sigma^2)^{-n/2-p/2-\alpha_0-1} \\
&\quad \cdot \exp \left\{ \frac{-1}{2\sigma^2} \left((\mathbf{y} - \mathbf{X}\boldsymbol{\theta})^T (\mathbf{y} - \mathbf{X}\boldsymbol{\theta}) + (\boldsymbol{\theta} - \boldsymbol{\mu}_0)^T \Sigma_0^{-1} (\boldsymbol{\theta} - \boldsymbol{\mu}_0) + 2\beta_0 \right) \right\}
\end{aligned}$$

(d) **Posterior Distribution.** Show that the posterior has the form

$$\mathbb{P}(\boldsymbol{\theta}, \sigma^2 | \mathbf{X}, \mathbf{y}) = \mathbb{P}(\boldsymbol{\theta} | \sigma^2, \mathbf{X}, \mathbf{y}) \cdot \mathbb{P}(\sigma^2, \mathbf{X}, \mathbf{y})$$

where

$$\begin{aligned}
\mathbb{P}(\boldsymbol{\theta}, \sigma^2, \mathbf{X}, \mathbf{y}) &= \mathcal{N}(\boldsymbol{\mu}_N, \sigma^2 \Sigma_N) \\
\mathbb{P}(\sigma^2 | \mathbf{X}, \mathbf{y}) &= \text{Inv-Gamma}(\alpha_N, \beta_N)
\end{aligned}$$

State the updated parameters.

Answer: First we isolate the terms relating to $\boldsymbol{\theta}$ found in part (c), call this $Q(\boldsymbol{\theta})$

$$\begin{aligned}
Q(\boldsymbol{\theta}) &= (\mathbf{y} - \mathbf{X}\boldsymbol{\theta})^T (\mathbf{y} - \mathbf{X}\boldsymbol{\theta}) + (\boldsymbol{\theta} - \boldsymbol{\mu}_0)^T \Sigma_0^{-1} (\boldsymbol{\theta} - \boldsymbol{\mu}_0) \\
&= \mathbf{y}^T \mathbf{y} - 2\boldsymbol{\theta}^T \mathbf{X}^T \mathbf{y} + \boldsymbol{\theta}^T \mathbf{X}^T \mathbf{X} \boldsymbol{\theta} + \boldsymbol{\theta}^T \Sigma_0^{-1} \boldsymbol{\theta} - 2\boldsymbol{\theta}^T \Sigma_0^{-1} \boldsymbol{\mu}_0 + \boldsymbol{\mu}_0^T \Sigma_0^{-1} \boldsymbol{\mu}_0 \\
&= \boldsymbol{\theta}^T \left(\mathbf{X}^T \mathbf{X} + \Sigma_0^{-1} \right) \boldsymbol{\theta} - 2\boldsymbol{\theta}^T \left(\mathbf{X}^T \mathbf{y} + \Sigma_0^{-1} \boldsymbol{\mu}_0 \right) + R
\end{aligned} \tag{1}$$

where R is the irrelevant terms not relating to $\boldsymbol{\theta}$. From (1), we can derive Σ_N^{-1} and $\boldsymbol{\mu}_N$ similar to how it was derived in the lectures. That is,

$$\begin{aligned}
\Sigma_N^{-1} &= \mathbf{X}^T \mathbf{X} + \Sigma_0^{-1} \\
\boldsymbol{\mu}_N &= \Sigma_N (\mathbf{X}^T \mathbf{y} + \Sigma_0^{-1} \boldsymbol{\mu}_0)
\end{aligned}$$

Subbing this result back into $\mathbb{P}(\boldsymbol{\theta})$, only concerning ourselves with terms that are relevant to $\boldsymbol{\theta}$ and σ^2 we get

$$\begin{aligned}
\mathbb{P}(\boldsymbol{\theta}, \sigma^2 | \mathbf{X}, \mathbf{y}) &= (\sigma^2)^{-n/2 - p/2 - \alpha_0 - 1} \exp \left\{ \frac{-1}{2\sigma^2} \left(\boldsymbol{\theta}^T \Sigma_N^{-1} \boldsymbol{\theta} - 2\boldsymbol{\theta}^T \Sigma_N^{-1} \boldsymbol{\mu}_N + R + 2\beta_0 \right) \right\} \\
&= (\sigma^2)^{-p/2} \exp \left\{ \frac{-1}{2\sigma^2} (\boldsymbol{\theta} - \boldsymbol{\mu}_N)^T \Sigma_N^{-1} (\boldsymbol{\theta} - \boldsymbol{\mu}_N) \right\}
\end{aligned} \tag{2}$$

$$\cdot (\sigma^2)^{-(n/2 + \alpha_0 + 1)} \exp \left\{ -\frac{\beta_0 + R/2}{\sigma^2} \right\} \tag{3}$$

Here, (2) = $\mathbb{P}(\boldsymbol{\theta} | \sigma^2, \mathbf{X}, \mathbf{y}) \propto \mathcal{N}(\boldsymbol{\mu}_N, \sigma^2 \Sigma_N)$ where

$$\begin{aligned}
\Sigma_N^{-1} &= \mathbf{X}^T \mathbf{X} + \Sigma_0^{-1} \\
\boldsymbol{\mu}_N &= \Sigma_N (\mathbf{X}^T \mathbf{y} + \Sigma_0^{-1} \boldsymbol{\mu}_0)
\end{aligned}$$

and (3) = $\mathbb{P}(\sigma^2, \mathbf{X}, \mathbf{y}) \propto \text{Inv-Gamma}(\alpha_N, \beta_N)$ where

$$\begin{aligned}
\alpha_N &= n/2 + \alpha_0 \\
\beta_N &= \beta_0 + R/2
\end{aligned}$$

Gradient of Logistic Regression Loss

Consider a binary classification problem with data

$$\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n, \quad \mathbf{x}_i \in \mathbb{R}^p, \quad y_i \in \{0, 1\}$$

Logistic regression models

$$h(\boldsymbol{\theta}) \triangleq \mathbb{P}(y = 1 | \mathbf{x}; \boldsymbol{\theta}) = \sigma(\langle \boldsymbol{\theta}, \mathbf{x} \rangle), \quad \sigma(z) = \frac{1}{1+e^{-z}}$$

The negative log-likelihood (cross-entropy loss) is

$$L(\boldsymbol{\theta}) = - \sum_{i=1}^n \left[y_i \log h_i(\boldsymbol{\theta}) + (1 - y_i) \log(1 - h_i(\boldsymbol{\theta})) \right],$$

where

$$h_i(\boldsymbol{\theta}) = \sigma(\langle \boldsymbol{\theta}, \mathbf{x}_i \rangle)$$

(a) Show that the derivative of the logistic function satisfies

$$\frac{d}{dz} \sigma(z) = \sigma(z)(1 - \sigma(z))$$

Answer: First evaluate LHS

$$\begin{aligned} \frac{d}{dz} \sigma(z) &= -1(1 + e^{-z})^{-2} \cdot -e^{-z} \\ &= \frac{e^{-z}}{(1 + e^{-z})^2} \end{aligned}$$

Now evaluate RHS

$$\begin{aligned} \sigma(z)(1 - \sigma(z)) &= \frac{1}{1 + e^{-z}} \left(1 - \frac{1}{1 + e^{-z}} \right) \\ &= \frac{1}{1 + e^{-z}} \left(\frac{1 + e^{-z} - 1}{1 + e^{-z}} \right) \\ &= \frac{e^{-z}}{(1 + e^{-z})^2} \\ &= \text{LHS} \end{aligned}$$

(b) Using the chain rule, derive the gradient of the loss function and show that

$$\nabla L(\boldsymbol{\theta}) = \sum_{i=1}^n (h_i(\boldsymbol{\theta}) - y_i) \mathbf{x}_i$$

Answer: Let's work with a single observation i . The cross-entropy loss for a single observation, $l_i(\boldsymbol{\theta})$ is given by

$$\begin{aligned} l_i(\boldsymbol{\theta}) &= -y_i \log h_i - (1 - y_i) \log(1 - h_i) \\ \frac{d}{dh_i(\boldsymbol{\theta})} l_i(\boldsymbol{\theta}) &= \frac{\partial_i}{\partial l_i} \cdot \frac{\partial h_i}{\partial \boldsymbol{\theta}} \\ &= \left(\frac{-y_i}{h_i} - \frac{(1 - y_i)}{(1 - h_i)} \right) \left(-1(1 - e^{-\langle \boldsymbol{\theta}, \mathbf{x}_i \rangle}) - \mathbf{x}_i e^{-\langle \boldsymbol{\theta}, \mathbf{x}_i \rangle} \right) \\ &= \frac{y_i(1 - h_i) - (1 - y_i)h_i}{h_i(1 - h_i)} \cdot h_i(1 - h_i) \mathbf{x}_i \\ &= -(y_i - h_i) \mathbf{x}_i \\ &= (h_i - y_i) \mathbf{x}_i \end{aligned}$$

Summing the cross-entropy loss for all observations we get

$$\nabla L(\boldsymbol{\theta}) = \sum_{i=1}^n (h_i - y_i) \mathbf{x}_i$$

(c) Write the gradient in matrix form

Answer: Looking at the answer from part (d), this can trivially be written in matrix form to

$$\nabla L(\boldsymbol{\theta}) = \mathbf{X}(\mathbf{h}(\boldsymbol{\theta}) - \mathbf{y})$$

2. Programming Exercise: Poisson Regression

The dataset `bikes.csv` contains daily bike rental counts. The variables are

- **temp**: temperature
- **humidity**: humidity level
- **windspeed**: wind speed
- **count**: number of bike rentals

We model the data using the Poisson regression model

$$y_i \sim \text{Pois}(\lambda_i), \quad \lambda_i = \exp(\langle \boldsymbol{\theta}, \mathbf{x}_i \rangle)$$

- Load the dataset and plot a histogram of **count**. Explain why a Poisson model may be appropriate
- Fit a Poisson regression model using the predictors (temp, humidity, windspeed)
- Interpret the coefficient of temp
- Use the fitted model to predict the expected number of rentals for

$$\text{temp} = 25, \text{humidity} = 50,$$

$$\text{windspeed} = 10$$

- Plot predicted counts versus observed counts.

Answer: See `w3q2.ipynb`